// EBRAHEIM MOHAMED PASHA QADRI

AI & Autonomous Systems Engineer

Fujairah, UAE | +971 50 690 7736 | ebraheimpasha@gmail.com | linkedin.com/in/ebraheim13ae

PROFESSIONAL SUMMARY

Computer and Autonomous Systems Engineering graduate specializing in AI-driven robotics and autonomous systems. Experienced in building multi-robot systems using ROS, SLAM, and computer vision, with additional industry exposure to PLC-based automation and sensor-driven control. Currently pursuing an MSc in Artificial Intelligence.

TECHNICAL SKILLS

Programming: Python, C++, MATLAB, Java

Robotics & Autonomous Systems: ROS, SLAM, TurtleBot3, Multi-Robot Systems, Task Allocation

AI & Computer Vision: OpenCV, CNN, PyTorch, Image Processing, Audio-Based Detection

Automation & Control: PLC Programming, Arduino, Raspberry Pi

Embedded Systems: VHDL, Vivado, Xilinx System Generator

Systems: Linux, MySQL, Networking Fundamentals

CERTIFICATIONS

Anthropic / Claude: AI Fluency: Framework & Foundations, AI Fluency for Builders, Claude Code 101, Claude Code in Action, Building with the Claude API, Introduction to Model Context Protocol (MCP), Claude Platform 101

EXPERIENCE

Backend AI Intern — FlyRank AI | *June 2026 – Ongoing*

- Designed API contracts and backend service logic supporting AI product features, with a focus on correctness and failure handling
- Built retrieval-backed answer flows (RAG) to ground AI responses in relevant source data
- Developed structured-output pipelines to produce reliable, schema-consistent responses from AI models

Engineering Intern — Protium Technologies, Ras Al Khaimah, UAE | *Jul 2025 – Jan 2026*

- Automated an algal bioreactor using AI-driven control logic for growth and nutrient optimization
- Designed decision-tree control logic using temperature, pH, and optical density sensors for real-time adjustments
- Developed and tested PLC code for motor control, valve operation, and data logging, supporting the transition from Arduino prototypes to industrial PLC systems
- Proposed and designed a biogas reactor system, including 3D design, component selection, and construction

PROJECTS

Gradfolio: AI-Powered Career Website Platform — Personal Project | *2026*

- Built and launched Gradfolio, a multi-user SaaS platform that converts CV data into interactive AI-powered career websites for students and fresh graduates
- Integrated Next.js, Supabase/PostgreSQL, Google Gemini, authentication, CV import, RLS security, and dynamic portfolio management into a production-deployed application
- Developed a verified-data AI career assistant and iterated on UX, responsive design, security, and deployment from internship brief to live product

AI SOC Triage Copilot Lite — Cybersecurity AI System | *2026*

- Developed an AI-based cybersecurity system to analyze and triage simulated security alerts through an end-to-end pipeline (alert input, feature extraction, AI-based reasoning, risk scoring, response generation)
- Implemented anomaly detection and contextual analysis to identify attack patterns across multiple events, integrating MITRE ATT&CK mapping to classify threats and explain attack techniques
- Built a reasoning layer to correlate logs and generate human-readable incident explanations, with automated generation of investigation steps and draft Sigma detection rules
- Simulated real-world SOC (Security Operations Center) workflows, designing the system to be scalable toward real-time monitoring and enterprise security environments

Swarm Robotics for Enhanced Indoor Safety Management — Thesis Project | *2024 – 2025*

- Developed a decentralized multi-robot system for autonomous indoor exploration using ROS and SLAM, with real-time map merging and adaptive navigation in unknown environments
- Designed a dynamic task allocation system using bidding and reassignment mechanisms for efficient multi-robot coordination
- Integrated CNN-based and audio-processing detection models to identify fire, vandalism, and distress events
- Validated performance in simulation and real-world environments, demonstrating improved efficiency over single-robot systems; research paper under review for publication

AI-Powered Autonomous Incident Detection & Response System — Personal Project | *2025*

- Developed an AI-driven system to monitor, analyze, and respond to real-time operational and security events through an end-to-end pipeline (event input, feature extraction, classification, automated response)
- Implemented machine learning models to classify event severity and prioritize response actions, combined with a hybrid rule-based/AI decision engine to trigger alerts, escalation, and mitigation actions
- Simulated real-world scenarios including abnormal behavior, system anomalies, and critical incidents, evaluating performance on response time, decision accuracy, and reliability

Agro Matic: Automated AI Hydroponics Farm System — Innovation Fair, UOWD | *2023 – 2024*

- Developed an AI-driven hydroponics system using IoT and machine learning to automate plant growth optimization
- Designed a system architecture integrating environmental sensors, control units, and a user interface for real-time monitoring of temperature, humidity, and nutrients

Car Black Box System — Innovation Fair (Early Engineering), UOWD | *2023*

- Developed an embedded safety system integrating multiple sensors for real-time accident and hazard detection
- Implemented automated emergency response logic, including alert systems and driver condition monitoring
- Achieved 3rd place at the innovation fair with a live demonstration

ADDITIONAL EXPERIENCE - VOLUNTEERING

Public Engagement & Protocol Ambassador — Expo 2020 DP World, Dubai | *Oct 2021 – Mar 2022*

- Coordinated and supervised a team of 10 staff members while supporting VIP protocol operations for high-profile delegates
- Demonstrated excellent communication across the management hierarchy, liaising effectively between frontline staff and senior pavilion leadership
- Supervised end-to-end execution of various events held at the pavilion

EDUCATION

MSc, Artificial Intelligence
Heriot-Watt University | Sep 2026 – Jul 2027 (Expected)

BEng, Computer and Autonomous Systems Engineering
University of Wollongong in Dubai | Completed 2026